

POWERBOX · DESIGN DECISION RECORD

PowerBox TRL 2 review decisions

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	PBX-DDR-001	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/powerbox		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Items 13 to 16 and 18 decided by Amish on 2026-10-02 (PBX-DEC-001)
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 decisions on the TRL 2 review and the move to SwapCell interface v0.3

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted (items 1 to 12; item 17 accepted by PBX-DDR-002); items 13 to 16 and 18 remained proposed at this record and were decided by Amish on 2026-10-02 as recommended in the design decisions register (PBX-DEC-001, items 3 and 5 to 8): "i approve your recommendations for all 555 open decisions."

■ Context

The TRL 2 review note (`docs/REVIEW.md` , sessions of 2026-09-24) listed the PowerBox design choices as "Proposed, awaiting Amish", most with a recommendation, and raised two interface issues with SwapCell (waking the pack and pass-through charging). On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." He also approved three cross-cutting additions to the SwapCell interface (a wake method for hosts without CAN, a charge-while-discharging mode and a latch vibration rating for vehicles), a rule that shared SwapCell packs are priced once and excluded from each dependent budget, and a rule that co-design partners are chosen per area later.

This record lists what that instruction decides and what it leaves open because there was no recommendation to accept. TRL 4 is on hold by Amish's instruction.

■ Options considered

The options for each item are in `docs/REVIEW.md` (2026-09-24) and PBX-PRC-001 v0.3. They are not repeated here.

■ Decision

Each item in Table 1 is "Decided by Amish, 2026-09-25: go with recommendation" unless the row says otherwise.

Table 1. Decided items.

#	Item	Decision	Where it now lives
1	Pack	Decided by Amish, 2026-09-25: go with recommendation. Removable SwapCell pack; the fixed 12.8 V LiFePO4 pack (about 384 Wh) stays only as a documented fallback	PBX-PRC-001 v0.4, README
2	Bay layout	Decided by Amish, 2026-09-25: go with recommendation. Horizontal side-loading bay with a door on the end	PBX-PRC-001 v0.4, <code>cad/src/model.py</code> , PBX-DWG-001
3	DC input	Decided by Amish, 2026-09-25: go with recommendation. One DC input with MPPT for a solar panel or SunSpoke's panel; a second input only if users need two sources at once	PBX-PRC-001 v0.4, PBX-REQ-001 R3
4	Grid charging	Decided by Amish, 2026-09-25: go with recommendation. Only through a certified external 54.6 V, 5 A charger brick; no AC inlet on the box	PBX-PRC-001 v0.4, PBX-REQ-001 R9
5	AC outlet	Decided by Amish, 2026-09-25: go with recommendation. Include the 300 W AC outlet with auto-off; document a DC-only variant (saves about \$92, no mains voltage)	PBX-PRC-001 v0.4
6	Enclosure	Decided by Amish, 2026-09-25: go with recommendation. Folded aluminium	PBX-PRC-001 v0.4, <code>bom/bom.csv</code>
7	AC region first	Decided by Amish, 2026-09-25: go with recommendation. 230 V 50 Hz with a 30 mA RCD; 120 V 60 Hz is a later variant	PBX-PRC-001 v0.4, PBX-REQ-001 R7, <code>bom/bom.csv</code> lines 9 and 12
8	Budget	Decided by Amish, 2026-09-25: go with recommendation. Keep <code>budget_usd: 450</code> ; it covers the PowerBox parts only. The SwapCell pack is priced once in the SwapCell BOM and excluded here (cross-cutting rule)	<code>project.yaml</code> (unchanged), PBX-REQ-001 R11, <code>bom/bom-notes.md</code>
9	Human-powered charging	Decided by Amish, 2026-09-25: go with recommendation. No human-powered input for PowerBox now; a pedal generator could later be a separate repo using the DC input	PBX-PRC-001 v0.4
10	Pass-through (station mode)	Decided by Amish, 2026-09-25 (cross-cutting approval): SwapCell adds a charge-while-discharging mode. PowerBox cites SwapCell interface v0.3 item C: host type 3 (station), requested mode 4 (charge-discharge), host charge limit in <code>HOST_HEARTBEAT</code>	PBX-PRC-001 v0.4, PBX-CAL-001 section 4
11	Waking the pack	Decided by Amish, 2026-09-25 (cross-cutting approval): SwapCell adds a wake method. PowerBox cites SwapCell interface v0.3 item W: every receiver fits a 10 k Ω coding resistor in its INTERLOCK loop instead of a direct link to ground; no keep-alive cell	PBX-PRC-001 v0.4, PBX-CAL-001 section 4, <code>bom/bom.csv</code> line 5
12	Latch class	Decided by Amish, 2026-09-25 (cross-cutting approval): SwapCell adds latch classes (item V). PowerBox is not a vehicle and uses a class D catch, with the bay door as a second stop; class V1 applies only if PowerBox is ever carried on a vehicle	PBX-PRC-001 v0.4

Items that remain open

These had no recommendation, or depend on users or on SwapCell, so they stayed **Proposed, awaiting Amish** at this record. All five were decided by Amish on 2026-10-02 (PBX-DEC-001). Item 17 had a recommendation and was decided on 2026-09-25 (PBX-DDR-002).

Table 2. Open items.

#	Item	Status
13	R2 shortfall (1.87 evenings against 2). Options: accept 1.87 evenings and relax R2; discharge to 3.5 % (a 5 % cut-off now gives only 1.97); or revise the reference profile with users (it must fall to about 189 Wh at the loads)	Decided by Amish, 2026-10-02 (PBX-DEC-001, item 5): R2 restated as 1.8 evenings (1.81 with minimum-capacity cells), the 10 % cut-off kept, the reference evening revisited with the first partner's users.
14	First co-design partner (NGO running community charging points, solar home system distributor, or university group)	Decided by Amish, 2026-10-02 (PBX-DEC-001, item 6): a partner in a 230 V, 50 Hz region that runs or supplies community charging or solar home systems; first candidate type to approach, a solar distribution charity such as SolarAid (Zambia and Malawi). Nothing is agreed.
15	Theft resistance: is a lockable pack door a priority? The door carries a padlock eye either way	Decided by Amish, 2026-10-02 (PBX-DEC-001, item 3): thumb-turn cam latch plus padlock hasp for the prototype; a keyed cam latch if the first partner runs an unattended or shared charging point.
16	Energy metering on the display for charging points that charge per phone	Decided by Amish, 2026-10-02 (PBX-DEC-001, item 7): energy and sessions counted per port in the host firmware and shown on the display, approximate and not for billing.
17	TRL 3 engineering proposals made in this session: a recessed normally closed wake button in series with the coding resistor, the 30 A buck, and the fuse ratings in PBX-CAL-001	Decided by Amish, 2026-09-25: go with recommendation (PBX-DDR-002)
18	Clarification requested from SwapCell: may a pack in legacy discharge (state 5) accept a station heartbeat and move to mode 2 or 4 without opening its output? PowerBox's host is powered from the pack and depends on this	Decided by Amish, 2026-10-02 (PBX-DEC-001, item 8): kept with SwapCell as one cross-repo action shared with MotionCore, answer needed before PowerBox's interface is frozen; if no, a small hold-up capacitor for the host.

■ Consequences

- PowerBox now builds to **SwapCell interface v0.3**. The receptacle must carry a 10 kΩ ±1 % INTERLOCK coding resistor; a direct link to SGND would leave the pack output dead.
- The two TRL 2 interface blockers (wake and pass-through) are closed on the SwapCell side. Item 18 is the one remaining interface question.
- The budget stays at \$450 and excludes the pack; the BOM totals \$449 (PBX-CAL-001), so R11 is at risk on indicative prices.
- R2 stays not met until item 13 is decided.